Technical review

Item #2 · foto-1.jpg · Run ID: 20261029T110057Z-95dba9

Path	`/tmp/tmp.sVTkOe7IbL/evidence/foto-1.jpg`
Size (bytes)	1483
Last modified (UTC)	2026-10-29T11:00:57Z
Extension	.jpg
Detected type (signature)	JPEG image
Status	Needs attention
Case	SAMPLE-1
Evidence ID	E0003
Case registry	Verified: identical to the registered evidence
Registered at intake (UTC)	2026-10-29T11:00:57.109Z

Hashes

sha256	`2c131e2942ef56ad7a4a6ecdce4740846e5c76f9d04fa46ac80b3753dad6b0cb`

Findings

[Low] Little or no camera metadata (METADATA_STRIPPED)

What it shows	The JPEG file carries no camera metadata (no make, model or capture date).
What it does not prove	Messengers, social networks, screenshots and privacy tools remove metadata routinely; its absence says nothing about the content.

[Informational] Standard (IJG) compression tables (QT_STANDARD_IJG)

What it shows	The quantization tables are the IJG standard tables at quality 85.
What it does not prove	Very many programs and devices use these tables; they identify a family of encoders, not one program or camera.

[Informational] Compression tables not in the database (QT_UNKNOWN)

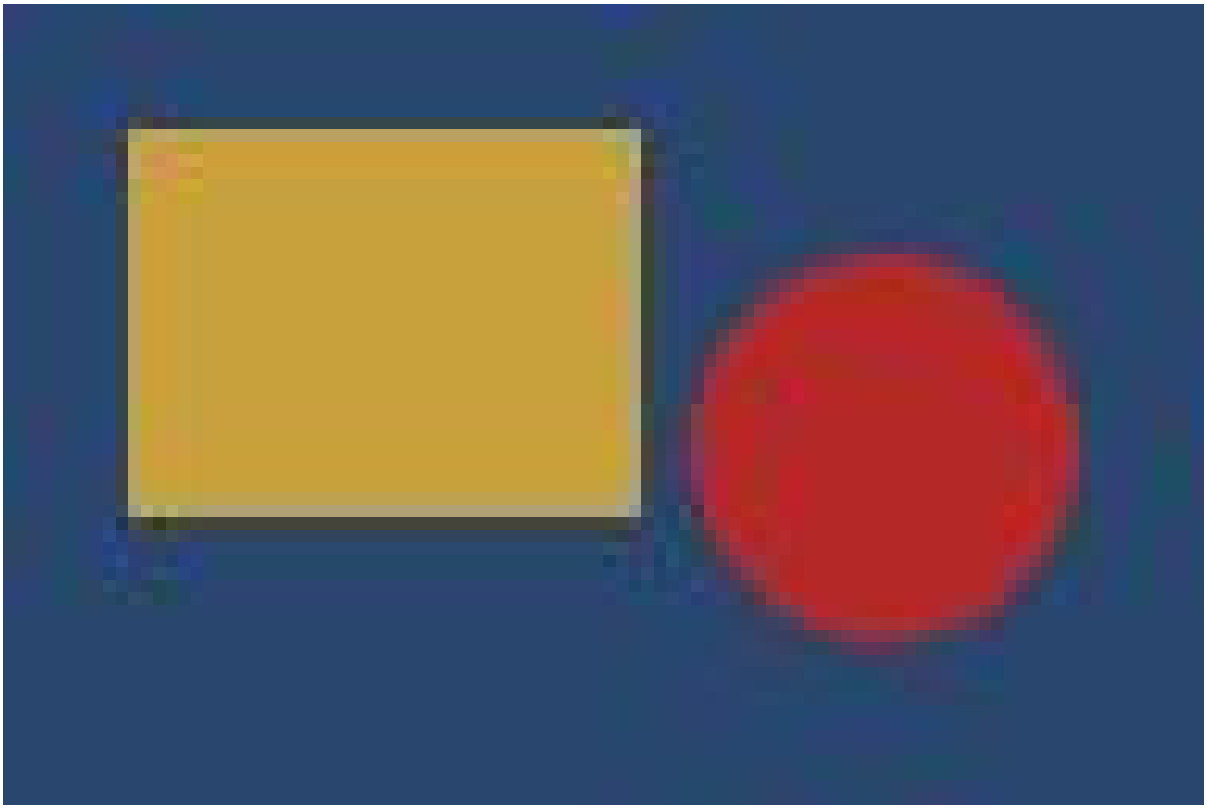
What it shows	No entry of the fingerprint databases matches these tables (SHA-256 43920728cfe021cebc3e0bf197ec4338c16eec09bf89e4b7a5cac6f67651ebfa).
What it does not prove	The database covers few devices and apps; an unknown table says nothing about authenticity. Add known originals with `fingerprints learn`.

Image analysis

Status	Completed
Format	JPEG
Dimensions (pixels)	96 × 64

Original image (decoded preview)

Lossless PNG of the decoded pixels, exactly as the analyses saw them (stored orientation; EXIF rotation not applied). The maps below have the same size and can be laid over it.



maps/original.png (96x64) — reduced copy for print; the full-resolution file is in the item folder

Metadata summary

Key fields only; metadata.json in the item folder holds every tag read. Metadata is written by the device or program that saved the file and can be edited or forged.

Tags read	0
Tags per group	—
GPS tags	absent
MakerNote	absent
XMP packet	absent
Embedded thumbnail	absent
ICC profile (SHA-256)	—

No metadata field was found.

Structure summary

Top-level segments in file order (JPEG markers, PNG chunks, RIFF chunks or HEIF boxes); structure.json holds the full tree with byte offsets.

Top-level segments	12
frame (SOF)	SOF0
chroma subsampling	4:2:0
components	3
restart interval	0
quantization tables	2
Huffman tables	4

bytes after EOI	0
comments (COM)	0

Marker / chunk / box sequence:

```
SOI · APP0/JFIF · DQT ×2 · SOF0 · DHT ×4 · SOS · EOI
```

Quantization tables

Values in natural (row-major) order, as stored in the DQT segments. The IJG comparison names a standard libjpeg quality only on an exact match; otherwise the nearest quality and its distance (sum of absolute differences) are given, which is not a quality percentage.

IJG comparison	IJG-standard q=85
Database match	no entry in the loaded databases
Table hash (SHA-256)	`43920728cfe021cebc3e0bf197ec4338c16eec09bf89e4b7a5cac6f67651ebfa`

Table 0 · 8-bit · Used by components: 1

```
5   3   3   5   7  12  15  18
4   4   4   6   8  17  18  17
4   4   5   7  12  17  21  17
4   5   7   9  15  26  24  19
5   7  11  17  20  33  31  23
7  11  17  19  24  31  34  28
15  19  23  26  31  36  36  30
22  28  29  29  34  30  31  30
```

Table 1 · 8-bit · Used by components: 2, 3

```
5   5   7  14  30  30  30  30
5   6   8  20  30  30  30  30
7   8  17  30  30  30  30  30
14  20  30  30  30  30  30  30
30  30  30  30  30  30  30  30
30  30  30  30  30  30  30  30
30  30  30  30  30  30  30  30
30  30  30  30  30  30  30  30
```

Fingerprint databases

Role	Database	Version	Entries	Status	SHA-256
starter	rakvor-starter	0	0	loaded	`e0b0c87e4b79e5c5402e920d343436056c25bb959a5bb757a7d0519b95ab4be8`
local	—	—	—	absent	—

Analysis maps

No analysis map was written for this item.

Copy-move shift vectors

Copy-move detection was not run (--copy-move off, or the pixels were not decoded).

Extracted files

structure.json (structure, 10889 B)	sha256: 41402f2c9c24d7b2ee0d2a46db75207d37612b7f0a126bca60baabf4f9d675c4
metadata.json (metadata, 123 B)	sha256: f07be8fa2c6e0057dc23806739e3091b649abb03e6cb9fe24458075adcbee13c

maps/original.png (original_preview, 4603 B)	sha256: 74b0017f5fb59449249ce61c3c350806a9e638c33ad79612f9f8396be883238a

Method

Each file was opened read-only and hashed in a single pass with the selected algorithms, and its real type was identified from its signature. JPEG, HEIC, PNG and WebP images were then re-read in an isolated process and the bytes read were hashed again and compared with the first pass. The image analysis read the file structure and metadata, compared the JPEG quantization tables with the IJG standard and the fingerprint databases, looked for double JPEG compression, wrote the selected visual maps (error level, noise, JPEG ghosts), searched for copied-and-moved regions, compared the embedded thumbnail with the main image and, when reference images were given, compared the image with them. HEIC pixels are not decoded (structure and metadata only). Every action and every file written was recorded in a hash-chained audit log.

Audit trail

Audit entries	#13 – #20
Log folder	`/tmp/tmp.sVTkOe7IbL/case/runs/rv-imgauth/20261029T110057Z-95dba9/logs`
Digital signature	This PDF is not digitally signed.

Limitations

- An image without findings is not proven authentic: careful edits can leave none of the traces these checks look for, and it is impossible to prove a negative (SWGDE 18-I-001).
- Error level, noise and ghost maps are visual aids for a trained examiner; the tool gives them no score and draws no conclusion from them.
- Double JPEG compression shows the image was saved more than once; ordinary sharing and exporting do this.
- The compression-table database covers few devices and apps (built from Rakvor's own captures plus local learning); an absent or matching entry is weak evidence on its own.
- Copy-move detection compares blocks at the same scale and orientation on a downscaled image: scaled, rotated or small copies are not found.
- Error rates of these checks have not yet been measured on a documented validation corpus; the findings are signals to be weighed by the examiner.
- Modification times come from the file system of the examined copy and may differ from the original source.
- A trusted timestamp proves the audit log existed in this exact form at the stated time; it does not prove that the examined copy matches the original source.
- Type detection uses a limited table of common file signatures; unknown types are reported as unknown, not as suspicious.